

Recap lecture 8

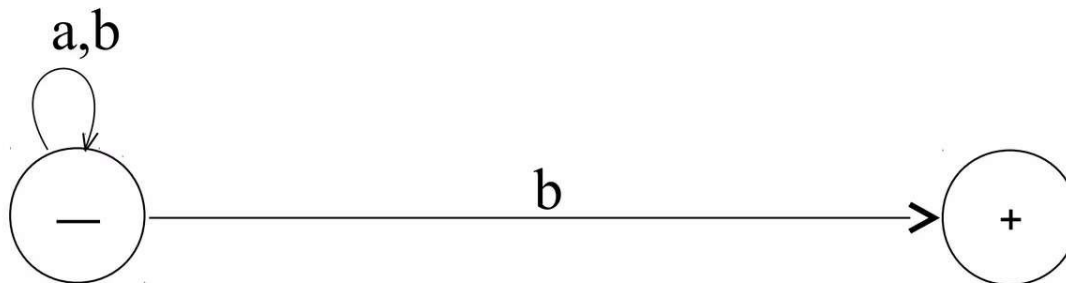


TG definition, Examples: accepting all strings, accepting none, starting with b, not ending in b, containing aa, containing aa or bb

Task Solution

Build a TG accepting the language L of strings, defined over $\Sigma=\{a, b\}$, **ending in b**.

Solution The language L may be expressed by RE $(a + b)^*b$, may be accepted by the following TG



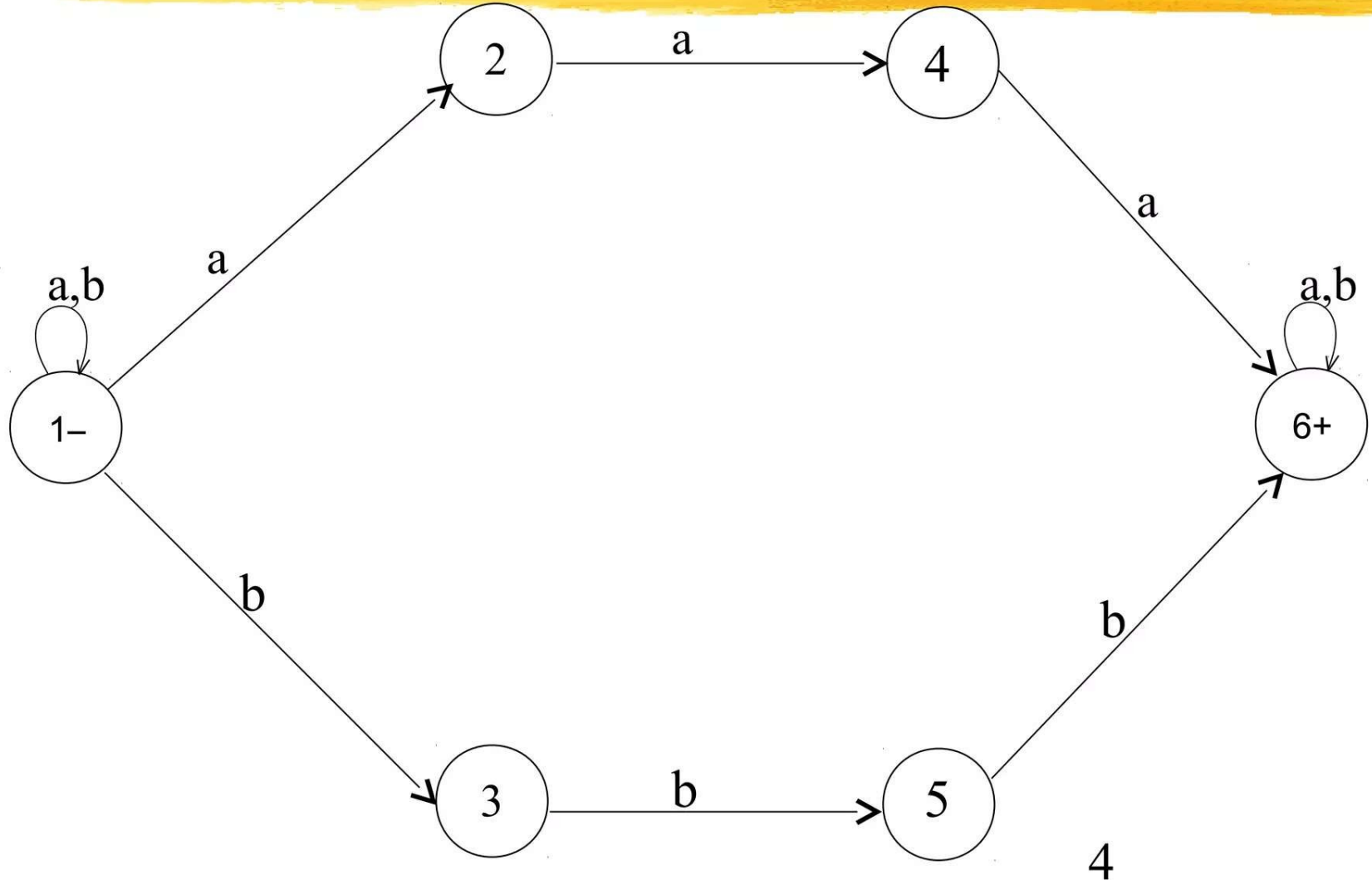
Example

Consider the language L of strings, defined over $\Sigma=\{a, b\}$, **having triple a or triple b**. The language L may be expressed by RE

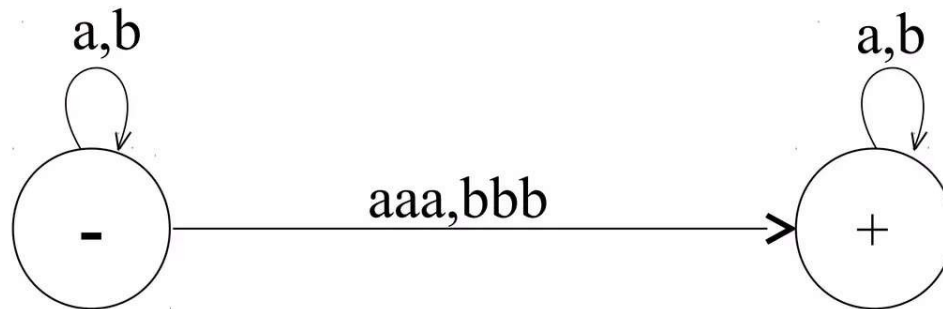
$$(a+b)^*(aaa + bbb)(a+b)^*$$

This language may be accepted by the following TG

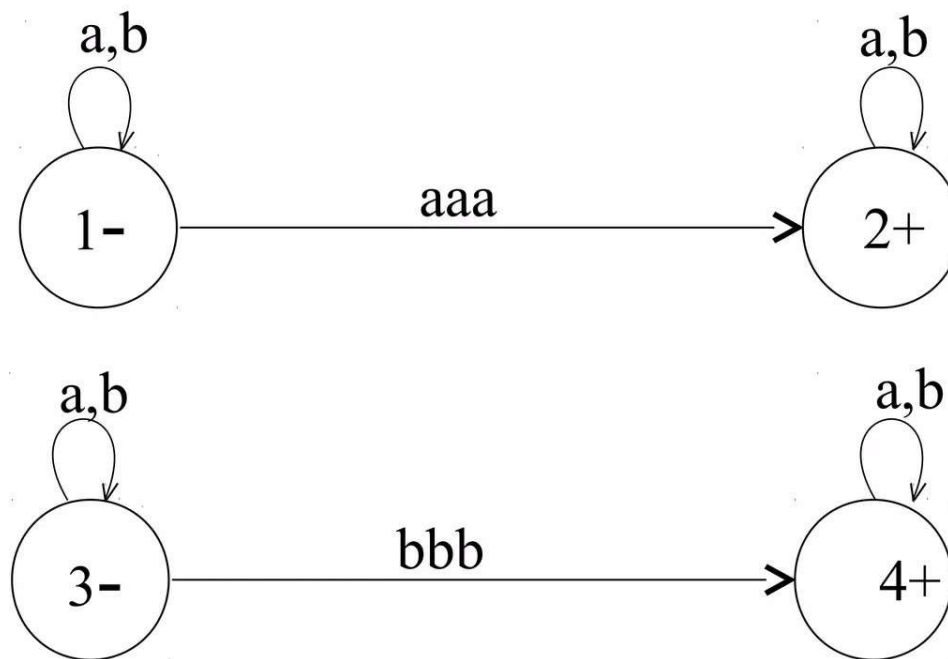
Example Continued ...



OR



OR



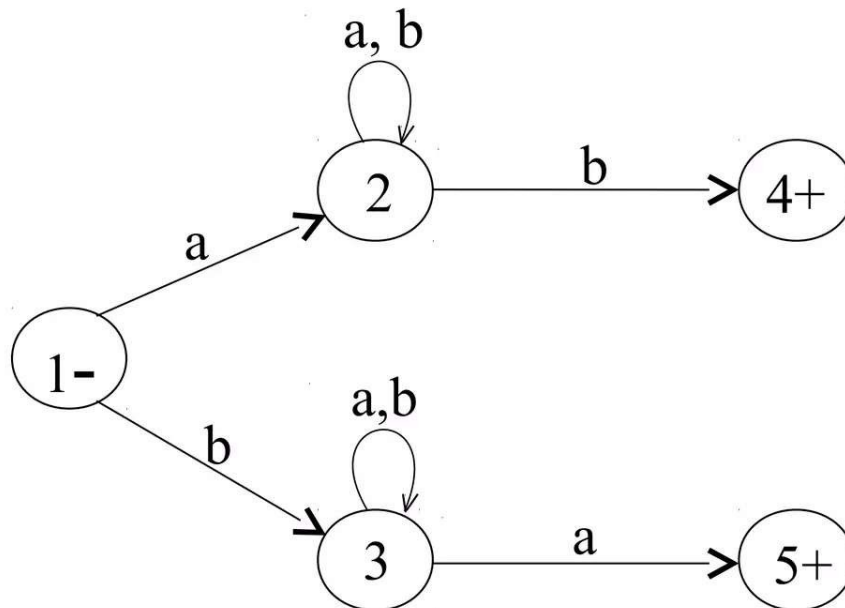
Example

Consider the language L of strings, defined over $\Sigma = \{a, b\}$, **beginning and ending in different letters.**

The language L may be expressed by RE
 $a(a + b)^*b + b(a + b)^*a$

The language L may be accepted by the following TG

Example continued ...



Example

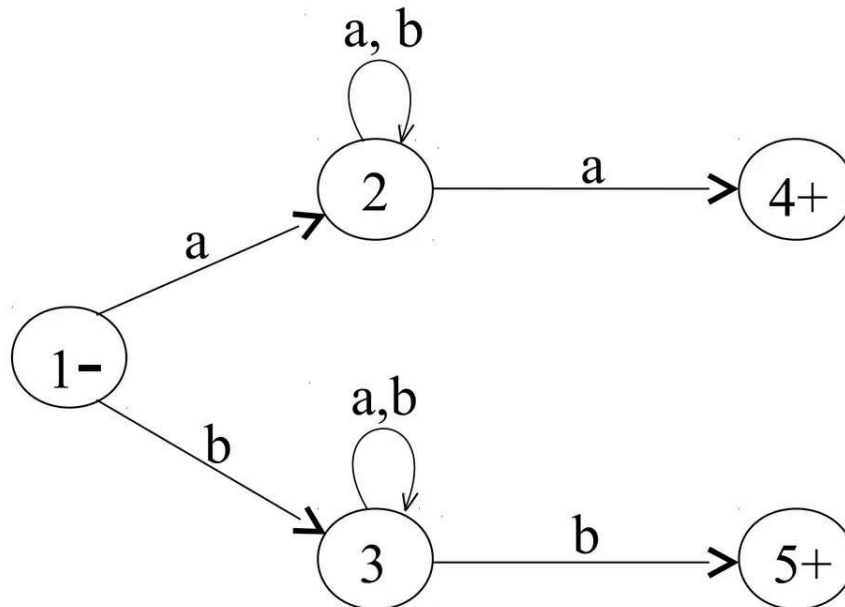
Consider the Language L of strings **of length two or more**, defined over $\Sigma = \{a, b\}$, **beginning with and ending in same letters.**

The language L may be expressed by the following regular expression

$$a(a + b)^*a + b(a + b)^*b$$

This language may be accepted by the following TG

Example Continued ...



Task



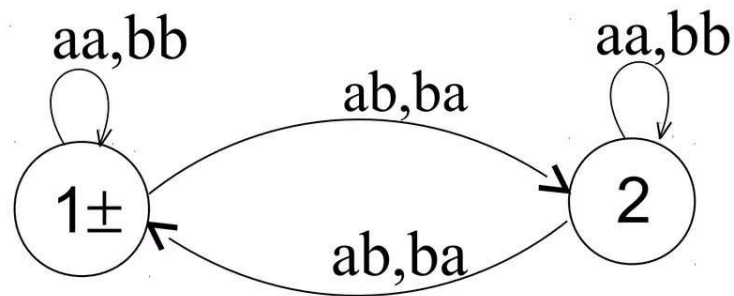
Build a TG accepting the language L of strings, defined over $\Sigma=\{a, b\}$, **beginning with and ending in the same letters.**

Example

Consider the **EVEN-EVEN** language, defined over $\Sigma=\{a, b\}$. As discussed earlier that **EVEN-EVEN** language can be expressed by a regular expression $(aa+bb+(ab+ba)(aa+bb)^*(ab+ba))^*$

The language **EVEN-EVEN** may be accepted by the following TG

Example continued ...



Example

Consider the language L , defined over $\Sigma=\{a, b\}$, in which **a's occur only in even clumps and that ends in three or more b's**. The language L can be expressed by its regular expression

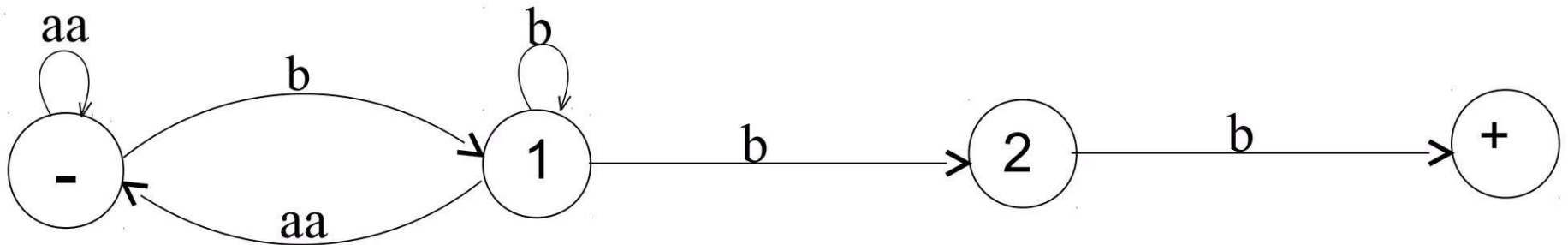
$$(aa)^*b(b^*+(aa(aa)^*b)^*)bb$$

OR

$$(aa)^*b(b^*+(aa)^*b)^*bb$$

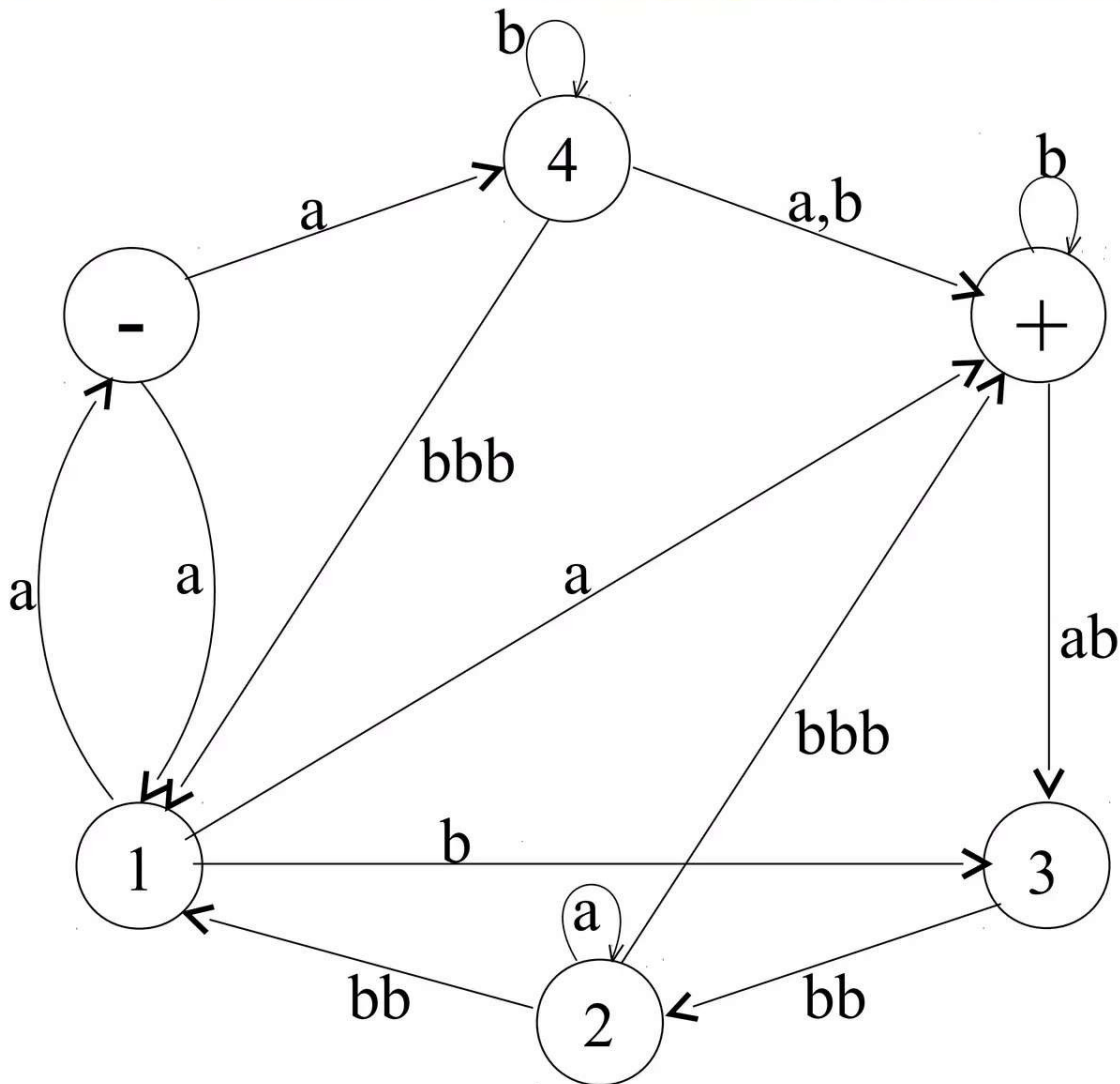
The language L may be accepted by the following TG

Example Continued ...



Example:

Consider the following TG



Example Continued ...

Consider the string abbbabbbabba. It may be observed that the above string traces the following three paths, (using the states)

- 1) (a)(b) (b) (b) (ab) (bb) (a) (bb) (a)
(-)(4)(4)(+)(+)(3)(2)(2)(1)(+)
- 2) (a)(b) ((b)(b)) (ab) (bb) (a) (bb) (a)
(-)(4)(+)(+)(+)(3)(2)(2)(1)(+)
- 3) (a) ((b) (b)) (b) (ab) (bb) (a) (bb) (a)
(-) (4)(4)(4)(+) (3)(2)(2)(1)(+)

Example Continued ...



Which shows that all these paths are successful, (*i.e.* the path starting from an initial state and ending in a final state).

Hence the string abbbabbbabba is accepted by the given TG.

Generalized Transition Graphs

A generalized transition graph (GTG) is a collection of three things

- 1) Finite number of states, at least one of which is start state and some (maybe none) final states.
- 2) Finite set of input letters (Σ) from which input strings are formed.
- 3) Directed edges connecting some pair of states labeled with regular expression.

It may be noted that in GTG, the labels of transition edges are corresponding regular expressions

Summing Up



**TGs accepting the languages:
containing aaa or bbb, beginning and
ending in different letters, beginning and
ending in same letters, EVEN-EVEN, a's
occur in even clumps and ends in three
or more b's, example showing different
paths traced by one string, Definition of
GTG**